Performance Estimation Problems (PEPs):

Tutorial – Session 2/3

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SMAI-MODE tutorial – March 2026

Main elements for today:

- ◇ What did we do so far? Recap' and takeaways,
- ◇ Search for structured convergence proofs.
- ◇ Search for structured "non-convergence" proofs (if time allows).
- ◇ Algorithm design.

Summary: key ingredients and observations

Towards structured performance analyses, take I

Towards structured performance analyses, take II

Concluding remarks

Example: analysis of a gradient method

Let $f : \mathbb{R}^d \to \mathbb{R}$ (continuously differentiable). Find $x_\star \in \mathbb{R}^d$ such that

$$f(x_\star) = \min_{x \in \mathbb{R}^d} f(x),$$

with f is L-smooth and μ-strongly convex ($f \in \mathcal{F}_{\mu,L}$).

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Question: what *a priori* guarantees after n iterations?

Examples: what about $f(x_n) - f(x_\star)$, $\|\nabla f(x_n)\|$, $\|x_n - x_\star\|$?

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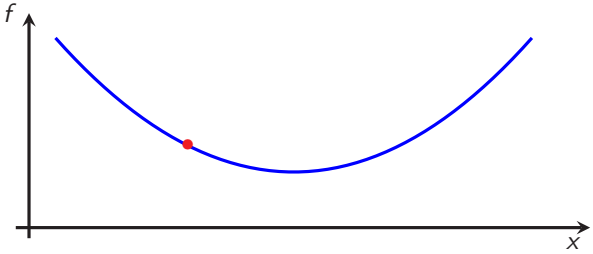

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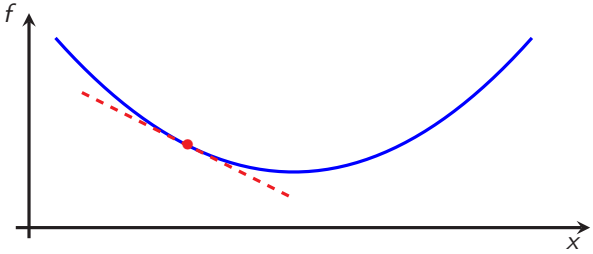

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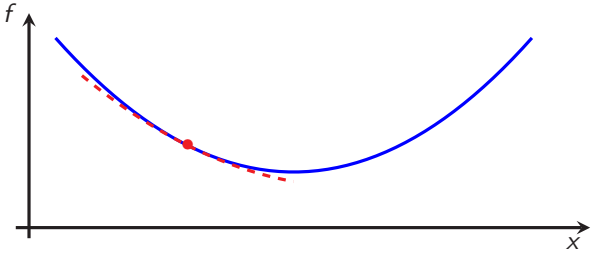

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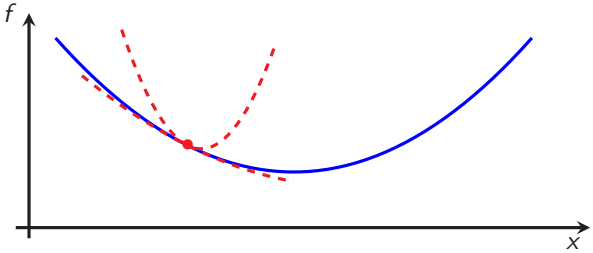

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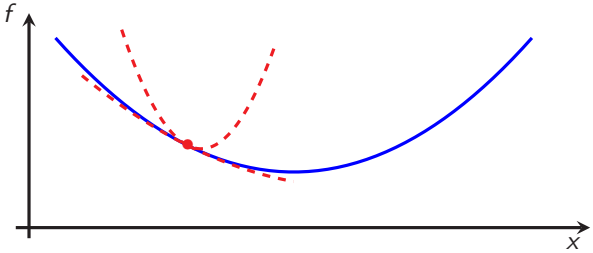

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(1&2) $\langle \nabla f(x) - \nabla f(y); x - y \rangle \geqslant \frac{1}{L+\mu}\|\nabla f(x) - \nabla f(y)\|^2 + \frac{\mu L}{L+\mu}\|x - y\|^2$.

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$$\Big\downarrow \text{ Inequality (1\&2)}$$

$$\leqslant \left(1 - \frac{2\alpha L \mu}{L+\mu}\right) \|x_0 - x_\star\|^2 + \alpha \left(\alpha - \frac{2}{L+\mu}\right) \|\nabla f(x_0)\|^2$$

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$$\begin{aligned}
\|x_1 - x_\star\|^2 &= \|x_0 - x_\star\|^2 - 2\alpha \langle \nabla f(x_0); x_0 - x_\star \rangle + \alpha^2 \|\nabla f(x_0)\|^2 \\
&\Big\downarrow \text{ Inequality (1\&2)} \\
&\leqslant \left(1 - \tfrac{2\alpha L \mu}{L+\mu}\right) \|x_0 - x_\star\|^2 + \alpha \left(\alpha - \tfrac{2}{L+\mu}\right) \|\nabla f(x_0)\|^2 \\
&\Big\downarrow \text{ if } 0 \leqslant \alpha \leqslant \tfrac{2}{L+\mu} \\
&\leqslant (1 - \alpha\mu)^2 \|x_0 - x_\star\|^2.
\end{aligned}$$

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- How to choose the right inequalities to combine?
- Why studying this specific quantity? Possible to adapt to other quantities?
- Unique way to arrive to the desired result?

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Legitimate questions:

- anything improvable? Realistic analyses?
- How to choose the right inequalities to combine?
- Why studying this specific quantity? Possible to adapt to other quantities?
- Unique way to arrive to the desired result?
- How likely are we to find such proofs in more complicated cases?

Legitimate questions about performance analyses?

Lemma 3. *Assume that the function is L-smooth and μ strongly-convex and satisfies the strong-growth condition in Equation 10. Then, using the updates in Equation 3-5 and setting the parameters according to Equations 17-19, if $\eta \leq \frac{1}{\mu L}$, then the following relation holds:*

$$\mathbb{E}_{k+1}\gamma_k^2 \left[\mathbb{E}\|\nabla f(w_{k+1}) - f^*\right] \leq \frac{\alpha_k^2}{\rho\eta}\left(f(x_0) - f^*\right) + \frac{\eta_k^2}{2\rho\eta}\|x_0 - w^*\|^2 + \frac{\sigma^2\eta}{\rho}\sum_{t=0}^{k}\left[\gamma_t^2\eta_{t+1}^2\right]$$

Proof.

Let $r_{k+1} = \|w_{k+1} - w^*\|$, then using equation 5

$$r_{k+1}^2 = \|\beta_k v_k + (1-\beta_k)\zeta_k - w^* - \gamma_k\eta\nabla f(\zeta_k, z_k)\|^2$$
$$r_{k+1}^2 = \|\beta_k v_k + (1-\beta_k)\zeta_k - w^*\|^2 + \gamma_k^2\eta^2\|\nabla f(\zeta_k, z_k)\|^2 + 2\gamma_k\eta\langle w^* - \beta_k v_k - (1-\beta_k)\zeta_k, \nabla f(\zeta_k, z_k)\rangle$$

Taking expectation wrt to z_k,

$$\begin{aligned}
\mathbb{E}[r_{k+1}^2] &= \mathbb{E}\|\beta_k v_k + (1-\beta_k)\zeta_k - w^*\|^2 + \gamma_k^2\eta^2\mathbb{E}\|\nabla f(\zeta_k, z_k)\|^2 + 2\gamma_k\eta\,\mathbb{E}[\langle w^* - \beta_k v_k - (1-\beta_k)\zeta_k, \nabla f(\zeta_k, z_k)\rangle] \\
&\leq \|\beta_k v_k + (1-\beta_k)\zeta_k - w^*\|^2 + \gamma_k^2\eta^2\rho\|\nabla f(\zeta_k)\|^2 + 2\gamma_k\eta\langle w^* - \beta_k v_k - (1-\beta_k)\zeta_k, \nabla f(\zeta_k)\rangle + \gamma_k^2\eta^2\sigma^2 \\
&= \|\beta_k(v_k - w^*) + (1-\beta_k)(\zeta_k - w^*)\|^2 + \gamma_k^2\eta^2\rho\|\nabla f(\zeta_k)\|^2 + 2\gamma_k\eta\langle(w^* - \beta_k v_k - (1-\beta_k)\zeta_k, \nabla f(\zeta_k)\rangle + \gamma_k^2\eta^2\sigma^2 \\
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&\qquad \text{(By convexity of } \|\cdot\|^2\text{)} \\
&= \beta_k r_k^2 + (1-\beta_k)\|\zeta_k - w^*\|^2 + \gamma_k^2\eta^2\rho\|\nabla f(\zeta_k)\|^2 + 2\gamma_k\eta\langle(w^* - \beta_k v_k - (1-\beta_k)\zeta_k, \nabla f(\zeta_k)\rangle + \gamma_k^2\eta^2\sigma^2 \\
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&\qquad \text{(From equation 4)} \\
&= \beta_k r_k^2 + (1-\beta_k)\|\zeta_k - w^*\|^2 + \gamma_k^2\eta^2\rho\|\nabla f(\zeta_k)\|^2 + 2\gamma_k\eta\left[\frac{\beta_k(1-\alpha_k)}{\alpha_k}\langle\nabla f(\zeta_k), (w_k - \zeta_k)\rangle + \langle\nabla f(\zeta_k), w^* - \zeta_k\rangle\right] + \gamma_k^2\eta^2\sigma^2 \\
&\leq \beta_k r_k^2 + (1-\beta_k)\|\zeta_k - w^*\|^2 + \gamma_k^2\eta^2\rho\|\nabla f(\zeta_k)\|^2 + 2\gamma_k\eta\left[\frac{\beta_k(1-\alpha_k)}{\alpha_k}(f(w_k) - f(\zeta_k)) + \langle\nabla f(\zeta_k), w^* - \zeta_k\rangle\right] + \gamma_k^2\eta^2\sigma^2 \\
&\qquad \text{(By convexity)}
\end{aligned}$$

By strong convexity,

$$\begin{aligned}
\mathbb{E}[r_{k+1}^2] &\leq \beta_k r_k^2 + (1-\beta_k)\|\zeta_k - w^*\|^2 + \gamma_k^2\eta^2\rho\|\nabla f(\zeta_k)\|^2 \\
&+ 2\gamma_k\eta\left[\frac{\beta_k(1-\alpha_k)}{\alpha_k}(f(w_k) - f(\zeta_k)) + f^* - f(\zeta_k) - \frac{\mu}{2}\|\zeta_k - w^*\|^2\right] + \gamma_k^2\eta^2\sigma^2
\end{aligned} \tag{16}$$

By Lipschitz continuity of the gradient,

$$\begin{aligned}
f(w_{k+1}) - f(\zeta_k) &\leq \langle\nabla f(\zeta_k), w_{k+1} - \zeta_k\rangle + \frac{L}{2}\|w_{k+1} - \zeta_k\|^2 \\
&\leq -\eta\langle\nabla f(\zeta_k), \nabla f(\zeta_k, z_k)\rangle + \frac{L\eta^2}{2}\|\nabla f(\zeta_k, z_k)\|^2
\end{aligned}$$

Taking expectation wrt z_k and using equations 9, 10

$$\mathbb{E}[f(w_{k+1}) - f(\zeta_k)] \leq -\eta\|\nabla f(\zeta_k)\|^2 + \frac{L\rho\eta^2}{2}\|\nabla f(\zeta_k)\|^2 + \frac{L\eta^2\sigma^2}{2}$$
$$\mathbb{E}[f(w_{k+1}) - f(\zeta_k)] \leq \left[-\eta + \frac{L\rho\eta^2}{2}\right]\|\nabla f(\zeta_k)\|^2 + \frac{L\eta^2\sigma^2}{2}$$

If $\eta \leq \frac{1}{\rho L}$,

$$\begin{aligned}
\mathbb{E}[f(w_{k+1}) - f(\zeta_k)] &\leq \left(\frac{-\eta}{2}\right)\|\nabla f(\zeta_k)\|^2 + \frac{L\eta^2\sigma^2}{2} \\
\implies \|\nabla f(\zeta_k)\|^2 &\leq \left(\frac{2}{\eta}\right)\mathbb{E}[f(\zeta_k) - f(w_{k+1})] + L\eta\sigma^2
\end{aligned} \tag{17}$$

From equations 16 and 17

$$\begin{aligned}
\mathbb{E}[r_{k+1}^2] &\leq \beta_k r_k^2 + (1-\beta_k)\|\zeta_k - w^*\|^2 + 2\gamma_k^2\eta\rho\mathbb{E}[f(\zeta_k) - f(w_{k+1})] \\
&+ 2\gamma_k\eta\left[\frac{\beta_k(1-\alpha_k)}{\alpha_k}(f(w_k) - f(\zeta_k)) + f^* - f(\zeta_k) - \frac{\mu}{2}\|\zeta_k - w^*\|^2\right] + \gamma_k^2\eta^2\sigma^2 + L\gamma_k^2\eta^2\rho\sigma^2 \\
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&= \beta_k r_k^2 + \|\zeta_k - w^*\|^2[(1-\beta_k) - \gamma_k\mu\eta] + f(\zeta_k)\left[2\gamma_k^2\eta\rho - 2\gamma_k\eta\cdot\frac{\beta_k(1-\alpha_k)}{\alpha_k} - 2\gamma_k\eta\right] \\
&- 2\gamma_k^2\eta\mathbb{E}f(w_{k+1}) + 2\gamma_k\eta f^* + \left[2\gamma_k\eta\cdot\frac{\beta_k(1-\alpha_k)}{\alpha_k}\right]f(w_k) + 2\gamma_k^2\eta^2\sigma^2
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$$\mathbb{E}_{k+1}\gamma_k^2 \left[\mathbb{E}f(w_{k+1}) - f^*\right] \leq \frac{\alpha_k^2}{\rho\eta}\left(f(\alpha_k) - f^*\right) + \frac{\mathbb{E}_k^2}{2\rho\eta}\left\|x_0 - w^*\right\|^2 + \frac{\sigma^2\eta}{\rho}\sum_{t=0}^{k}\left[\gamma_t^2 \mathbb{E}_{t+1}^2\right]$$

Proof.

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$$r_{k+1}^2 = \|\beta_k v_k + (1-\beta_k)\zeta_k - w^* - \gamma_k \eta \nabla f(\zeta_k, z_k)\|^2$$
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&\qquad \text{(By convexity of } \|\cdot\|^2\text{)} \\
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&= \beta_k r_k^2 + (1-\beta_k)\|\zeta_k - w^*\|^2 + \gamma_k^2\eta^2\rho\|\nabla f(\zeta_k)\|^2 + 2\gamma_k\eta\langle\beta_k(\zeta_k - v_k) + w^* - \zeta_k, \nabla f(\zeta_k)\rangle + \gamma_k^2\eta^2\sigma^2 \\
&= \beta_k r_k^2 + (1-\beta_k)\|\zeta_k - w^*\|^2 + \gamma_k^2\eta^2\rho\|\nabla f(\zeta_k)\|^2 + 2\gamma_k\eta\left\langle\frac{\beta_k(1-\alpha_k)}{\alpha_k}(w_k - \zeta_k) + w^* - \zeta_k, \nabla f(\zeta_k)\right\rangle + \gamma_k^2\eta^2\sigma^2 \\
&\qquad \text{(From equation 4)} \\
&= \beta_k r_k^2 + (1-\beta_k)\|\zeta_k - w^*\|^2 + \gamma_k^2\eta^2\rho\|\nabla f(\zeta_k)\|^2 + 2\gamma_k\eta\left[\frac{\beta_k(1-\alpha_k)}{\alpha_k}\langle\nabla f(\zeta_k), (w_k - \zeta_k)\rangle + \langle\nabla f(\zeta_k), w^* - \zeta_k\rangle\right] + \gamma_k^2\eta^2\sigma^2 \\
&\leq \beta_k r_k^2 + (1-\beta_k)\|\zeta_k - w^*\|^2 + \gamma_k^2\eta^2\rho\|\nabla f(\zeta_k)\|^2 + 2\gamma_k\eta\left[\frac{\beta_k(1-\alpha_k)}{\alpha_k}(f(w_k) - f(\zeta_k)) + \langle\nabla f(\zeta_k), w^* - \zeta_k\rangle\right] + \gamma_k^2\eta^2\sigma^2 \\
&\qquad \text{(By convexity)}
\end{aligned}$$

By strong convexity,

$$\mathbb{E}[r_{k+1}^2] \leq \beta_k r_k^2 + (1-\beta_k)\|\zeta_k - w^*\|^2 + \gamma_k^2\eta^2\rho\|\nabla f(\zeta_k)\|^2 + 2\gamma_k\eta\left[\frac{\beta_k(1-\alpha_k)}{\alpha_k}(f(w_k) - f(\zeta_k)) + f^* - f(\zeta_k) - \frac{\mu}{2}\|\zeta_k - w^*\|^2\right] + \gamma_k^2\eta^2\sigma^2 \tag{16}$$

By Lipschitz continuity of the gradient,

$$f(w_{k+1}) - f(\zeta_k) \leq \langle\nabla f(\zeta_k), w_{k+1} - \zeta_k\rangle + \frac{L}{2}\|w_{k+1} - \zeta_k\|^2$$
$$\leq -\eta\langle\nabla f(\zeta_k), \nabla f(\zeta_k, z_k)\rangle + \frac{L\eta^2}{2}\|\nabla f(\zeta_k, z_k)\|^2$$

Taking expectation wrt z_k and using equations 9, 10

$$\mathbb{E}[f(w_{k+1}) - f(\zeta_k)] \leq -\eta\|\nabla f(\zeta_k)\|^2 + \frac{L\rho\eta^2}{2}\|\nabla f(\zeta_k)\|^2 + \frac{L\eta^2\sigma^2}{2}$$
$$\mathbb{E}[f(w_{k+1}) - f(\zeta_k)] \leq \left[-\eta + \frac{L\rho\eta^2}{2}\right]\|\nabla f(\zeta_k)\|^2 + \frac{L\eta^2\sigma^2}{2}$$

If $\eta \leq \frac{1}{\mu L}$,

$$\mathbb{E}[f(w_{k+1}) - f(\zeta_k)] \leq \left(\frac{-\eta}{2}\right)\|\nabla f(\zeta_k)\|^2 + \frac{L\eta^2\sigma^2}{2}$$
$$\implies \|\nabla f(\zeta_k)\|^2 \leq \left(\frac{2}{\eta}\right)\mathbb{E}[f(\zeta_k) - f(w_{k+1})] + L\eta\sigma^2 \tag{17}$$

From equations 16 and 17

$$\begin{aligned}
\mathbb{E}[r_{k+1}^2] &\leq \beta_k r_k^2 + (1-\beta_k)\|\zeta_k - w^*\|^2 + 2\gamma_k^2\eta\rho\mathbb{E}[f(\zeta_k) - f(w_{k+1})] \\
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&= \beta_k r_k^2 + \|\zeta_k - w^*\|^2\left[(1-\beta_k) - \gamma_k\mu\eta\right] + f(\zeta_k)\left[2\gamma_k^2\eta\rho - 2\gamma_k\eta\cdot\frac{\beta_k(1-\alpha_k)}{\alpha_k} - 2\gamma_k\eta\right] \\
&\quad - 2\gamma_k^2\eta\mathbb{E}f(w_{k+1}) + 2\gamma_k\eta f^* + \left[2\gamma_k\eta\cdot\frac{\beta_k(1-\alpha_k)}{\alpha_k}\right]f(w_k) + 2\gamma_k^2\eta^2\sigma^2
\end{aligned}$$

Example - do not read!
Example - do not read!

- ⚠ Error-prone
- ⚠ Technical, lack global insights.
- ⚠ Few proof patterns.

- ? Easily fixable?
- ? Simple to adapt to variations of target inequality?
- ? Simple to adapt to algorithmic variations?

Performance estimation: key ingredients & summary

- Cast worst-case analysis problem as optimization problem
  - worst-case with respect to function to be minimized $\rightarrow$ infinite dimensional.
- Analysis problem depends on function only at iterates and optimal point:
  - $\rightarrow$ analysis problem can be reformulated using the *interpolation trick*.
  - ☺ interpolation conditions are known for *simple* classical sets of functions.
- Problem turns out to be linear in Gram matrix and function values
  - $\rightarrow$ semidefinite lifting.
- Leverage dual problem (also an SDP) to search for proofs
  - $\rightarrow$ proofs $\equiv$ weighted sums of quadratic inequalities,
  - $\rightarrow$ such proofs can be converted to "standard proofs".

Convergence rate of a gradient step

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Toy example: What is the smallest τ such that:

$$\|x_1 - x_\star\|^2 \leqslant \tau \|x_0 - x_\star\|^2,$$

for all

- $d \in \mathbb{N}$, L-smooth and μ-strongly convex function f (notation $f \in \mathcal{F}_{\mu,L}$),
- x_0, and x_1 generated by gradient step $x_1 = x_0 - \alpha \nabla f(x_0)$,
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$$\tau = \max_{f, \mathbf{x_0}, \mathbf{x_1}, x_\star} \frac{\|x_1 - x_\star\|^2}{\|x_0 - x_\star\|^2}$$

$$\text{s.t. } f \in \mathcal{F}_{\mu,L}$$

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$$\begin{aligned} \text{s.t. } & f \in \mathcal{F}_{\mu,L} && \text{Functional class} \\ & x_1 = x_0 - \alpha \nabla f(x_0) && \text{Algorithm} \\ & \nabla f(x_\star) = 0 && \text{Optimality of } x_\star \end{aligned}$$

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<u>Variables</u>: f, x_0, x_1, $x_\star$.

<u>Parameters</u>: μ, L, α.

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Semidefinite formulation

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- For $d = 1$ same as original problem by adding $\operatorname{rank}(G) \leqslant 1$.

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Fix $L = 1$, $\mu = .1$ and solve the SDP for a few values of α.

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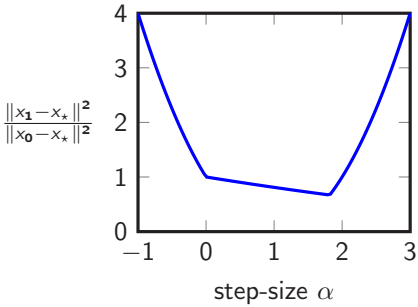

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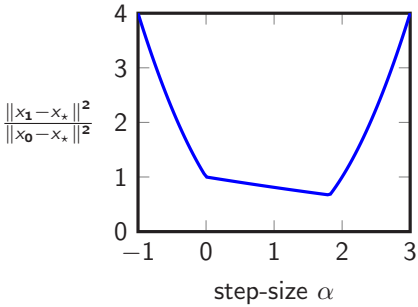

- Observation: numerics match $\max\{(1-\alpha L)^2, (1-\alpha\mu)^2\}$.

Dual problem

- Dual problem is

$$\min_{\tau,\lambda_1,\lambda_2 \geqslant 0} \tau$$

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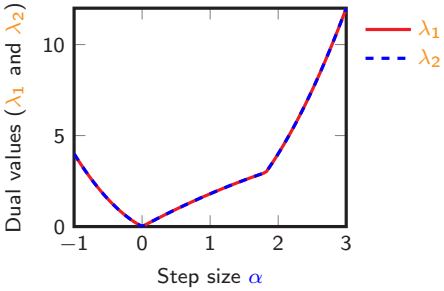

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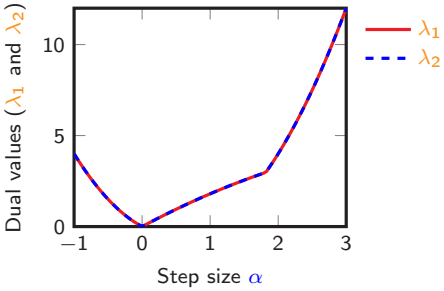

Numerics match $\lambda_1 = \lambda_2 = 2|\alpha|\rho(\alpha)$ with $\rho(h) = \max\{\alpha L - 1, 1 - \alpha\mu\}$.

Recovering a "standard" proof

Gradient with $\alpha = \frac{1}{L}$. Perform weighted sum of two inequalities

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What can we guarantee on...

$$\frac{f(x_n) - f(x_\star)}{\|x_0 - x_\star\|^2} \leqslant ? \qquad \frac{\|\nabla f(x_n)\|^2}{\|x_0 - x_\star\|^2} \leqslant ? \qquad \frac{\min_{0 \leqslant k \leqslant n} \|\nabla f(x_k)\|^2}{\|x_0 - x_\star\|^2} \leqslant ?$$

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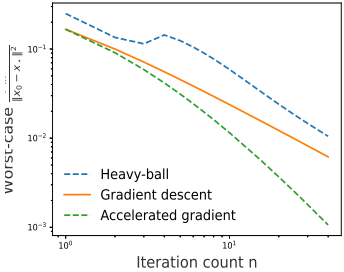

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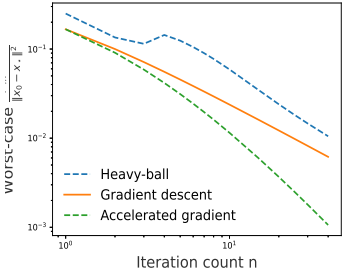

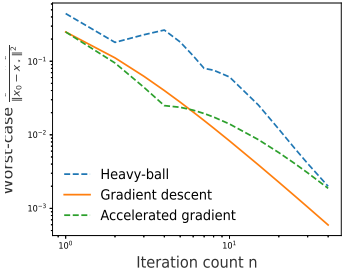

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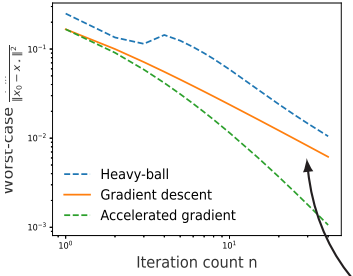

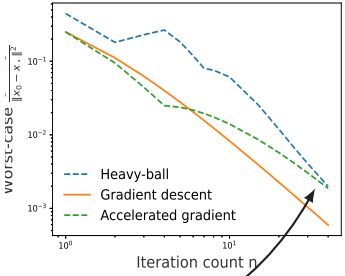

What to expect for larger n?

Summary: key ingredients and observations

Towards structured performance analyses, take I

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Concluding remarks

Back to gradient descent

Let $f : \mathbb{R}^d \to \mathbb{R}$ (continuously differentiable). Find $x_\star \in \mathbb{R}^d$ such that

$$f(x_\star) = \min_{x \in \mathbb{R}^d} f(x),$$

with f is L-smooth and μ-strongly convex ($f \in \mathcal{F}_{\mu,L}$).

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Question: what *a priori* convergence rate?

Examples: we have seen $\frac{\|x_1 - x_\star\|^2}{\|x_0 - x_\star\|^2}$, what about $\frac{\|\nabla f(x_1)\|^2}{\|\nabla f(x_0)\|^2}$, $\frac{f(x_1) - f_\star}{f(x_0) - f_\star}$?

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$$\frac{a\|x_1 - x_\star\|^2 + b\|\nabla f(x_1)\|^2}{a\|x_0 - x_\star\|^2 + b\|\nabla f(x_0)\|^2}.$$

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- What about optimizing over it?

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- More details: pick $F=[f_0,\, f_1]$ ($f_\star=0$, $x_\star=0$ wlog) and

$$G = \begin{pmatrix} \|x_0\|^2 & \langle g_0; x_0\rangle & \langle g_1; x_0\rangle \\ \langle g_0; x_0\rangle & \|g_0\|^2 & \langle g_1; g_0\rangle \\ \langle g_1; x_0\rangle & \langle g_1; g_0\rangle & \|g_1\|^2 \end{pmatrix}.$$

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$$\max_{G\succcurlyeq 0, F} \operatorname{Tr}(C_{\text{obj}}G)$$

$$\begin{aligned} \text{s.t. interp.: } & \operatorname{Tr}(M_{ij}G)+\langle v_{ij}; F\rangle \leqslant 0, \\ \text{init.: } & \operatorname{Tr}(C_{\text{init}}G)\leqslant 1. \end{aligned}$$

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Define <u>selectors</u> for G:

$\bar{x}_0 \triangleq e_1,\; \bar{g}_0 \triangleq e_2,\; \bar{g}_1 \triangleq e_3, \bar{x}_1 \triangleq \bar{x}_0-\alpha\bar{g}_0.$

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- More details: pick $F=[f_0,\, f_1]$ ($f_\star=0$, $x_\star=0$ wlog) and

$$G = \begin{pmatrix} \|x_0\|^2 & \langle g_0; x_0\rangle & \langle g_1; x_0\rangle \\ \langle g_0; x_0\rangle & \|g_0\|^2 & \langle g_1; g_0\rangle \\ \langle g_1; x_0\rangle & \langle g_1; g_0\rangle & \|g_1\|^2 \end{pmatrix}.$$

- Resulting semidefinite program:

$$\begin{aligned} \max_{G\succcurlyeq 0,F} \; & \mathrm{Tr}(C_{\mathrm{obj}}G) \\ \text{s.t. interp.: } & \mathrm{Tr}(M_{ij}G)+\langle v_{ij}; F\rangle \leqslant 0, \\ \text{init.: } & \mathrm{Tr}(C_{\mathrm{init}}G)\leqslant 1. \end{aligned}$$

Define <u>selectors</u> for G:

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Back to smooth strongly convex problems

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- Observe: fixed $\rho \Rightarrow$ constraints are linear in (a, b).
- So: for any fixed $\rho > 0$: verifying if $\exists (a, b)$ such that

$$\frac{a\|x_1 - x_\star\|^2 + b\|\nabla f(x_1)\|^2}{a\|x_0 - x_\star\|^2 + b\|\nabla f(x_0)\|^2} \leqslant \rho$$

(for all $f \in \mathcal{F}_{\mu,L},\ x_\star, x_0, x_1 = x_0 - \alpha \nabla f(x_0)$) is convex problem.

Example: inexact gradient method

Setup: minimize $f \in \mathcal{F}_{\mu,L}$ using

$$x_{k+1} = x_k - \alpha d_k$$

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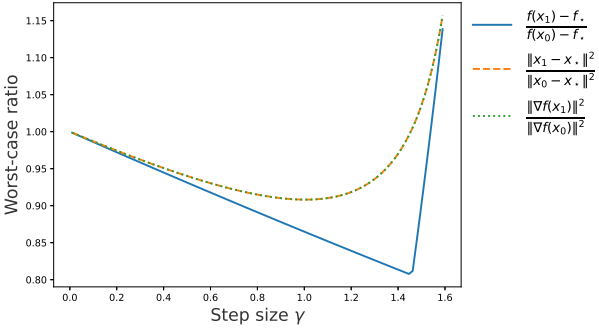

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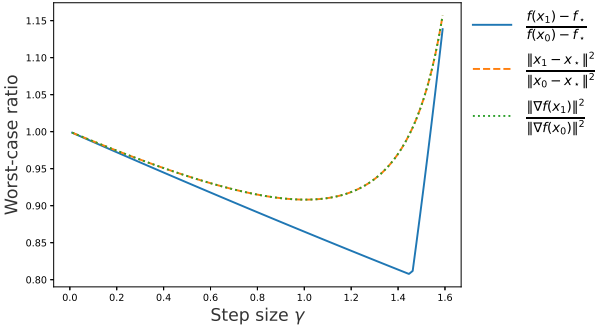

$\rightarrow$ picking the right metric affects the conclusion, even in simple cases!

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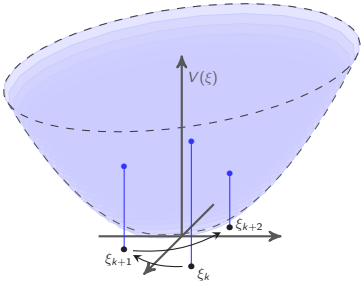

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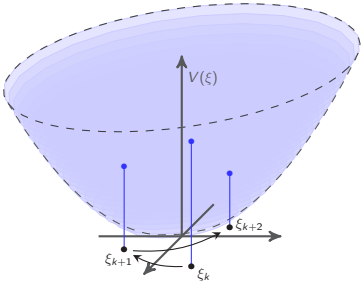

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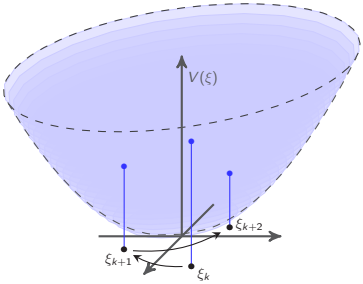

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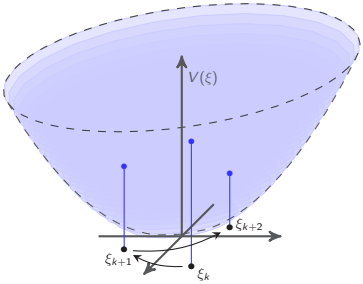

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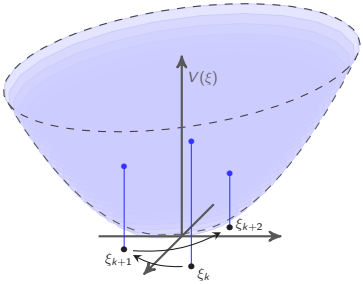

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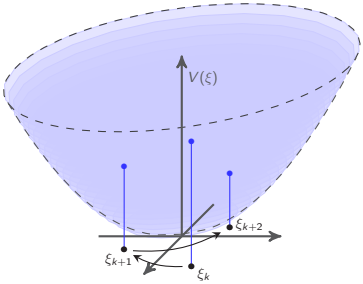

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A Lyapunov family for gradient descent

Gradient descent: $x_{k+1} = x_k - \alpha \nabla f(x_k)$. Reasonable choice:

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Verify first property

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- for which $V(\xi) \geqslant \|x - x_\star\|^2$ for all $d \in \mathbb{N}$, $\xi \in (\mathbb{R}^d)^2 \times (\mathbb{R})^2$, $f \in \mathcal{F}_{\mu,L}$, and
- for which $V(\xi_{k+1}) \leqslant \rho V(\xi_k)$ for all $d \in \mathbb{N}$, $\xi_k \in (\mathbb{R}^d)^2 \times (\mathbb{R})^2$, $f \in \mathcal{F}_{\mu,L}$, compatible ξ_{k+1}.

Verify first property $\Leftrightarrow \forall \epsilon \geqslant 0$:

$$\epsilon \leqslant \inf_{\substack{d \in \mathbb{N}, f \in \mathcal{F}_{\mu,L} \\ x, x_\star \in \mathbb{R}^d}} V\left([x - x_\star, \nabla f(x), f(x) - f(x_\star)]\right)$$

$$\text{s.t. } \|x - x_\star\|^2 \geqslant \epsilon,$$

$$\nabla f(x_\star) = 0,$$

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Optimization variables: $d, f, x, x_\star$ (includes optimization over space $f \in \mathcal{F}_{\mu,L}$).

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$\Rightarrow$ verification of conditions can be reframed as LMIs (bonus: linear in $(P, p) \in \mathbb{S}^2 \times \mathbb{R}$).

Positive definite Lyapunov candidate for gradient descent

Interpolation conditions allow removing red constraints

$$\epsilon \leqslant \inf_{\substack{d\in\mathbb{N}, f_x, f_\star \in \mathbb{R} \\ x, x_\star, g_x, g_\star \in \mathbb{R}^d}} V\left([x - x_\star, g_x, f_x - f_\star]\right)$$

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replacing them by

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with $g_\star = 0$.

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with $g_\star = 0$. Same optimal value (no relaxation).

Positive definite Lyapunov candidate for gradient descent

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- Using new variables $G \succcurlyeq 0$ and F

$$G = \begin{bmatrix} \|x - x_\star\|^2 & \langle g_x, x - x_\star \rangle \\ \langle g_x, x - x_\star \rangle & \|g_x\|_2^2 \end{bmatrix}, \quad F = f_x - f_\star,$$

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- Homogeneous in $\epsilon \Rightarrow$ we can fix $\epsilon = 1$ wlog.
- No duality gap (e.g., $\exists$ Slater point).

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- via Lagrange dual:

$$\epsilon \leqslant \sup_{\tau, \lambda_1^{\mathrm{Lyap}}, \lambda_2^{\mathrm{Lyap}} \geqslant 0} \tau\epsilon$$

$$P + \lambda_1^{\mathrm{Lyap}} A_1 + \lambda_2^{\mathrm{Lyap}} A_2 \succcurlyeq \tau \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix},$$

$$p - \lambda_1^{\mathrm{Lyap}} + \lambda_2^{\mathrm{Lyap}} \geqslant 0.$$

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$\rightarrow$ LMI (linear in $(P, p, \lambda_1^{\mathsf{Lyap}}, \lambda_2^{\mathsf{Lyap}})$) by fixing $\tau = 1$ (fixing the scale of the Lyapunov).

Positive definite Lyapunov candidate for gradient descent

... in other words, our candidate Lyapunov function

$$V(\xi) = \begin{pmatrix} x - x_\star \\ g_x \end{pmatrix}^T (P \otimes I_d) \begin{pmatrix} x - x_\star \\ g_x \end{pmatrix} + p^T (f_x - f_\star)$$

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Methods with memory – same story

$$x_{k+1} = x_k - \alpha \nabla f(x_k) - \beta \nabla f(x_{k-1}) + \gamma (x_k - x_{k-1}).$$

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Ultimately, searching Lyapunov functions via two LMIs:

1. verifying $V(\xi) \geqslant \|x - x_\star\|^2$ for all $d \in \mathbb{N}$, $\xi \in (\mathbb{R}^d)^4 \times (\mathbb{R})^2$, $f \in \mathcal{F}_{\mu,L}$,
2. verifying $V(\xi_{k+1}) \leqslant \rho V(\xi_k)$ for all $d \in \mathbb{N}$, $\xi_k \in (\mathbb{R}^d)^4 \times (\mathbb{R})^2$, $f \in \mathcal{F}_{\mu,L}$, compatible ξ_{k+1}.

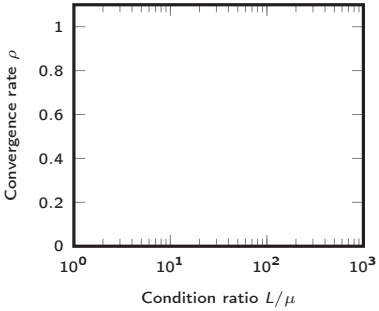
... and bisection search for optimizing ρ (quasi-convex problem).

Examples: stationary first-order methods

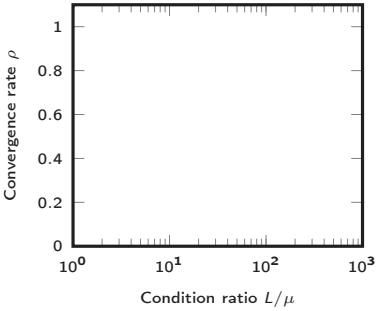
Method	α	β	γ

$$x_k = y_k + \gamma(y_k - y_{k-1})$$
$$y_{k+1} = y_k + \beta(y_k - y_{k-1}) - \alpha\nabla f(x_k)$$

"Lyapunov rates"



"Lyapunov rates" with $P \succcurlyeq 0$, $p \geqslant 0$



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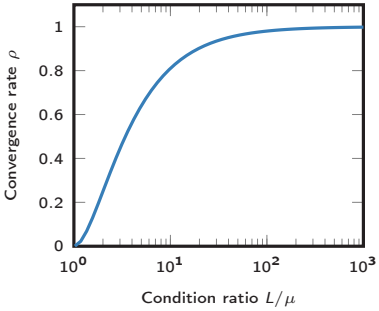

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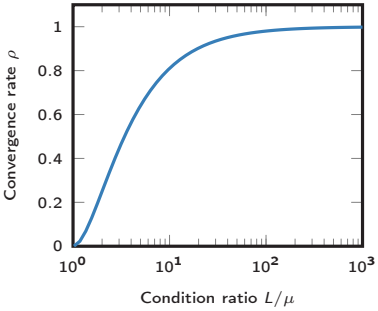

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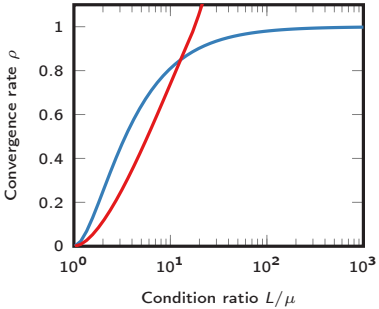
Polyak's	$\frac{4}{(\sqrt{L}+\sqrt{\mu})^2}$	$\left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)^2$	0

"Lyapunov rates"



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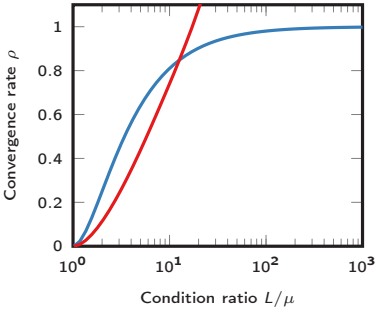

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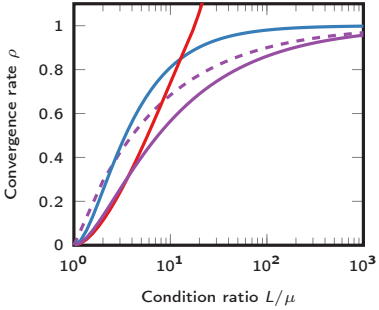

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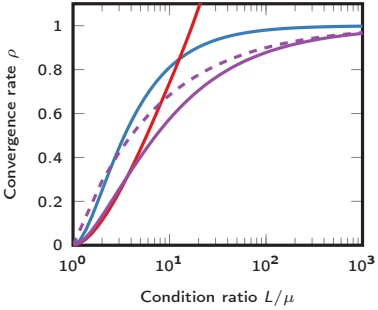

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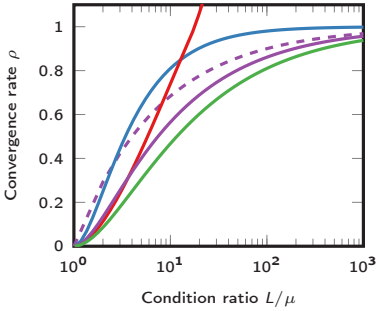
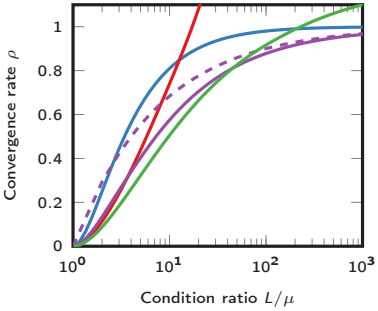
Van Scoy's	$\frac{2\sqrt{L}-\sqrt{\mu}}{L\sqrt{L}}$	$\frac{(\sqrt{\kappa}-1)^2}{\kappa+\sqrt{\kappa}}$	$\frac{(\sqrt{\kappa}-1)^2}{2\kappa+\sqrt{\kappa}-1}$

"Lyapunov rates"



"Lyapunov rates" with $P \succcurlyeq 0$, $p \geqslant 0$



Cycles

Heavy-ball

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Pick specific (α, β) and fix cycle length K.

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Look for non-trivial cycles of length $K \in \mathbb{N}$ by solving:

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Cycles

Heavy-ball

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From same steps as before $\rightarrow$ SDP formulation $\rightarrow$ LP (via convexity and symmetries).

Heavy-ball method: Lyapunov vs. cycles

Heavy-ball: $x_{k+1} = x_k - \alpha \nabla f(x_k) + \beta(x_k - x_{k-1})$. Choices of (α, β) for convergence?[5,6,7]

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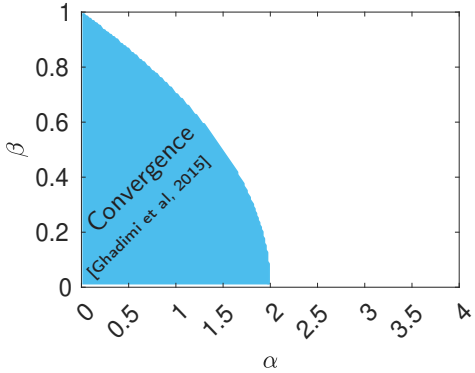

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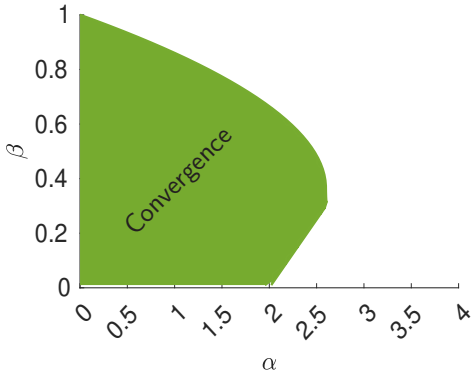

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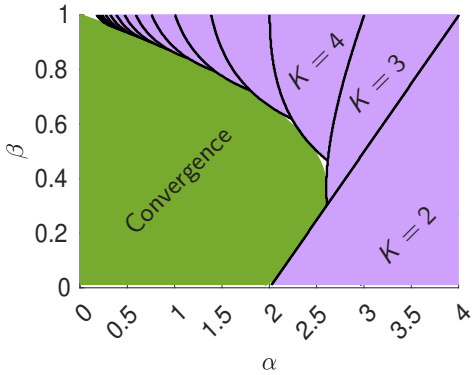

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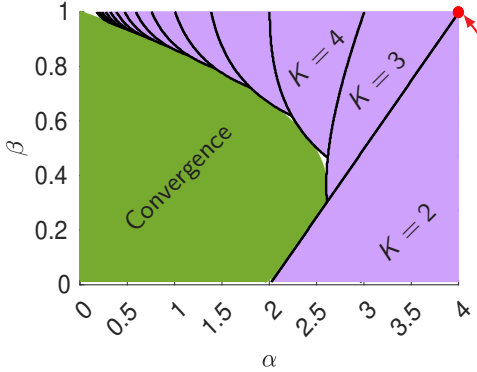

"Optimal tuning" for quadratic optimization

(numerics for $L/\mu = 10^7$)

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Summary: key ingredients and observations

Towards structured performance analyses, take I

Towards structured performance analyses, take II

Concluding remarks

Lyapunov functions for gradient descent

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Not new... (Nesterov 1983), (Beck & Teboulle 2009), (Bansal & Gupta 2017), (Wilson, Recht & Jordan 2016) and many others.

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Examples: analyses of a gradient method via potential functions

- ☺ advantages of Lyapunov analyses (simplicity)
- ☹ we loose tightness guarantees from PEPs, but ...
- ☺ we know exactly what we loose (use solution to PEPs as references),
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How to choose a_k, b_k, c_k, d_k's?

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Notation: set of (ϕ_k, ϕ_{k+1}) such that $\phi_{k+1} \leqslant \phi_k$ (for all compatible f, x_k, x_{k+1}, $x_\star$) is denoted $\mathcal{V}_k$.

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Let's reverse-engineer a proof:

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Motivation: this structure gives $f(x_n) - f(x_\star) \leqslant \frac{L\,\|x_0 - x_\star\|^2}{2d_n}$. Best achievable guarantee such proofs?

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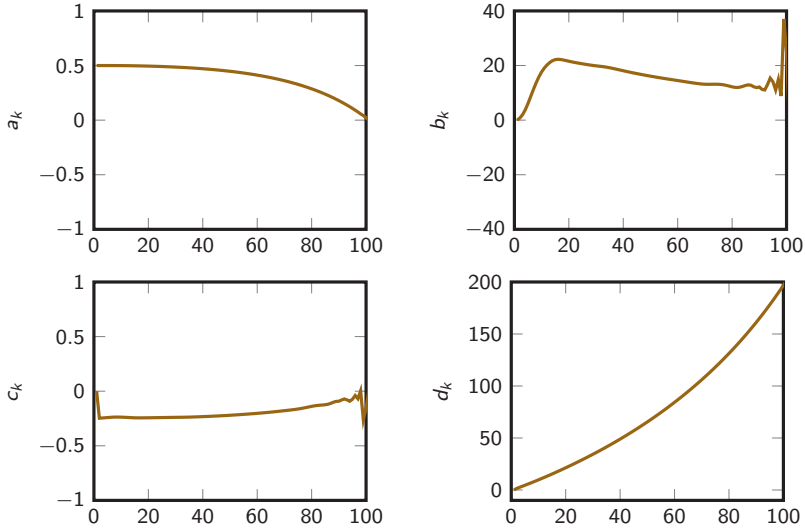

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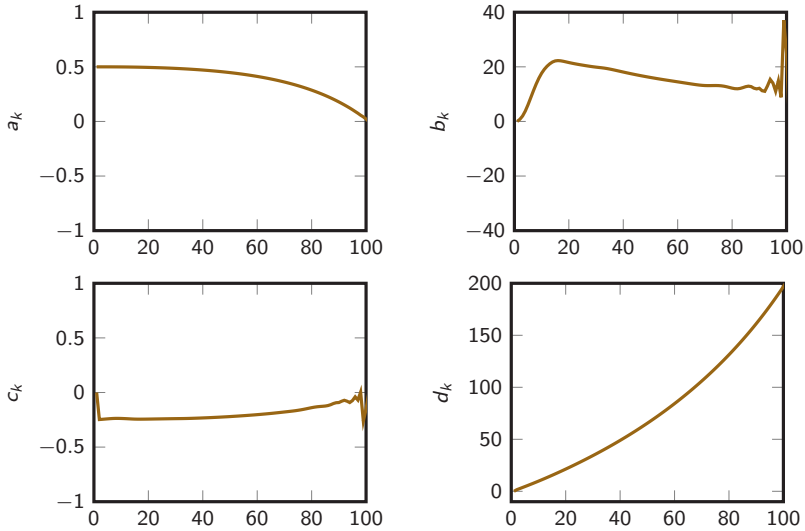

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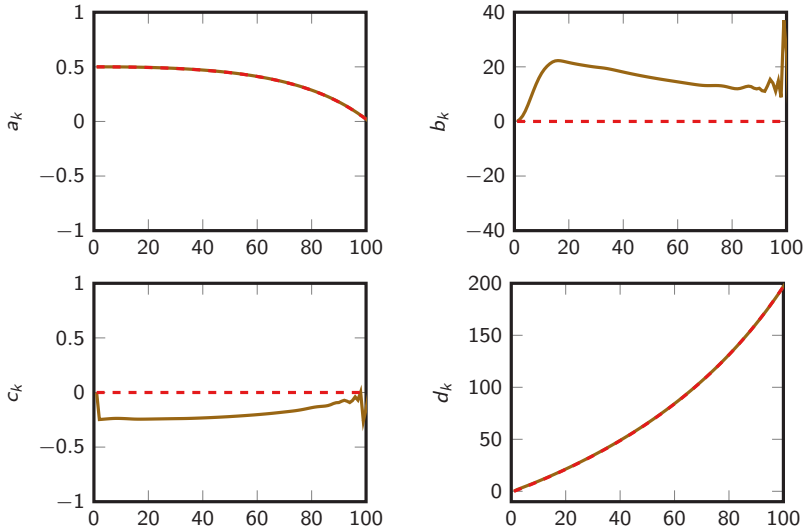

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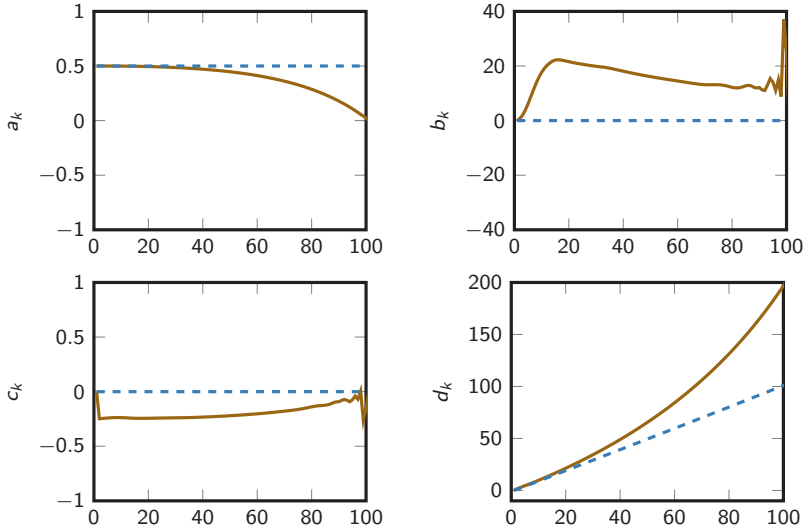

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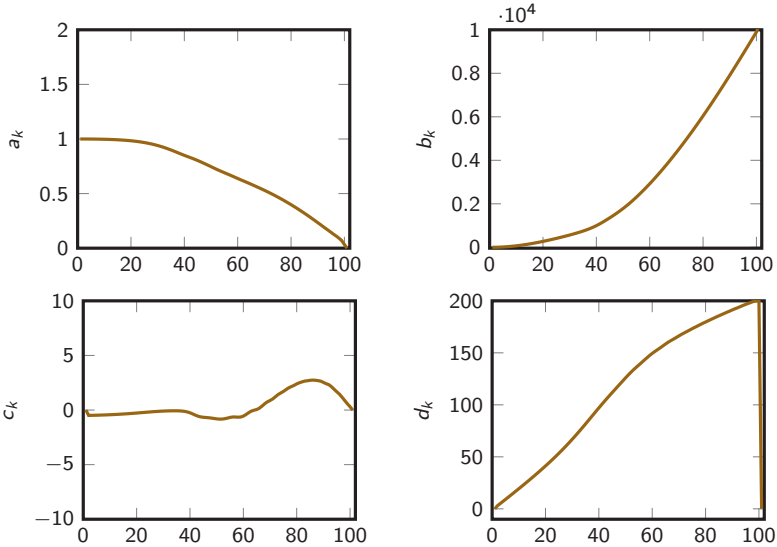

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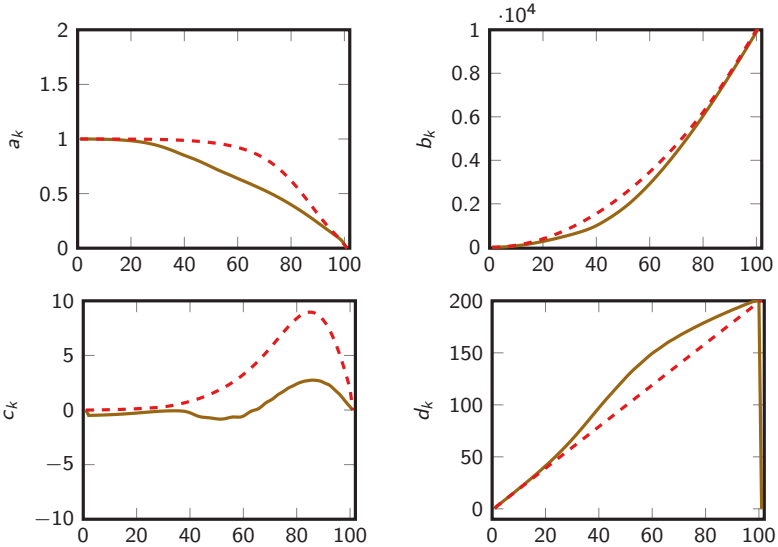

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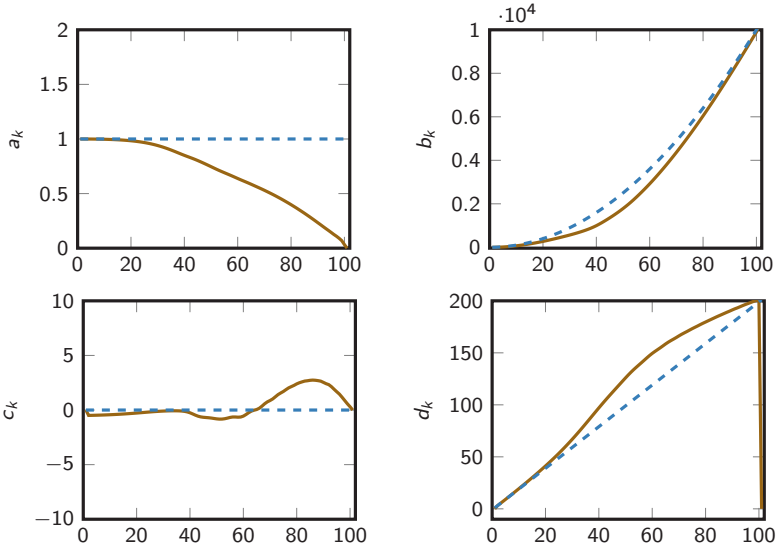

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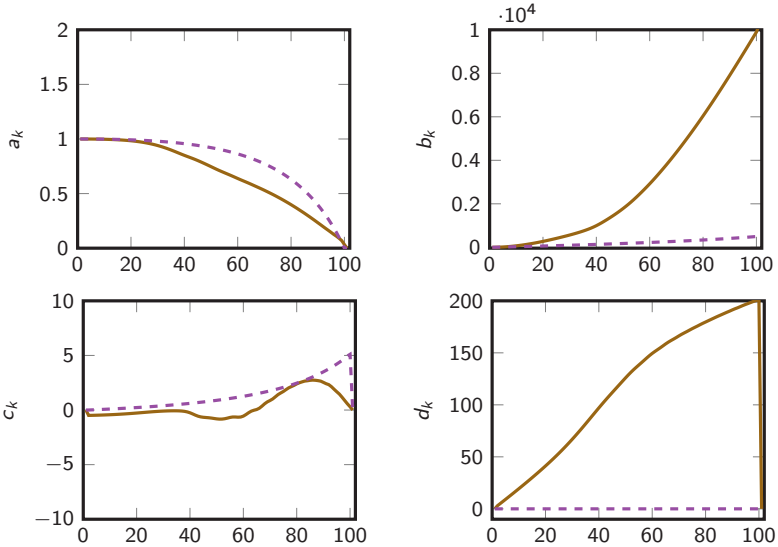

Summary: key ingredients and observations

Towards structured performance analyses, take I

Towards structured performance analyses, take II

Concluding remarks

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- ☺/☹ Proofs generally not unique.

Heuristic considerations

What is a simple proof?

What's next?

What we did so far:

- worst-case analyses from PEP perspective,
- proof structures from PEP perspective.
- Restrict (proof) search space to certain (simpler) structures.

What's next?

- *Practical* proof design using PEPs and Lyapunov functions.
- How to leverage all those tools to perform actual algorithm design?